



Biometrics Final Project

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Project Overview

Biometric:

Speaker Recognition



System Process:

1. Sound Sample Collection
2. Preprocessing
3. Transformer Feature Extraction Model
4. Similarity Scoring
5. Decision Implementation

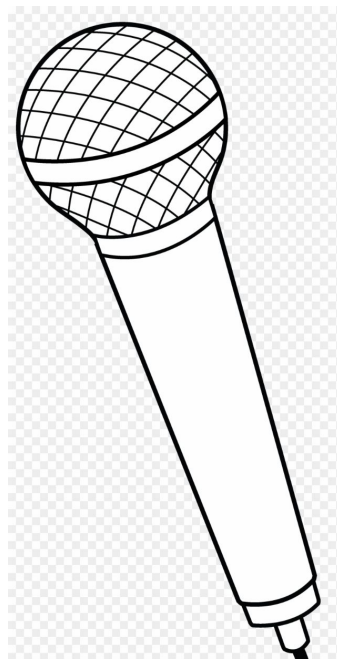
Fun Application:

Smart Toy Nerf Blaster



Sample Collection

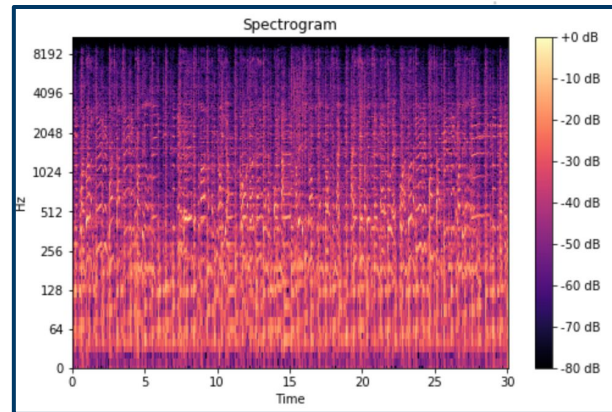
Our system collects sound directly from the computer microphone, which is where all the processing is done.



Preprocessing (HomeGrown)

Steps:

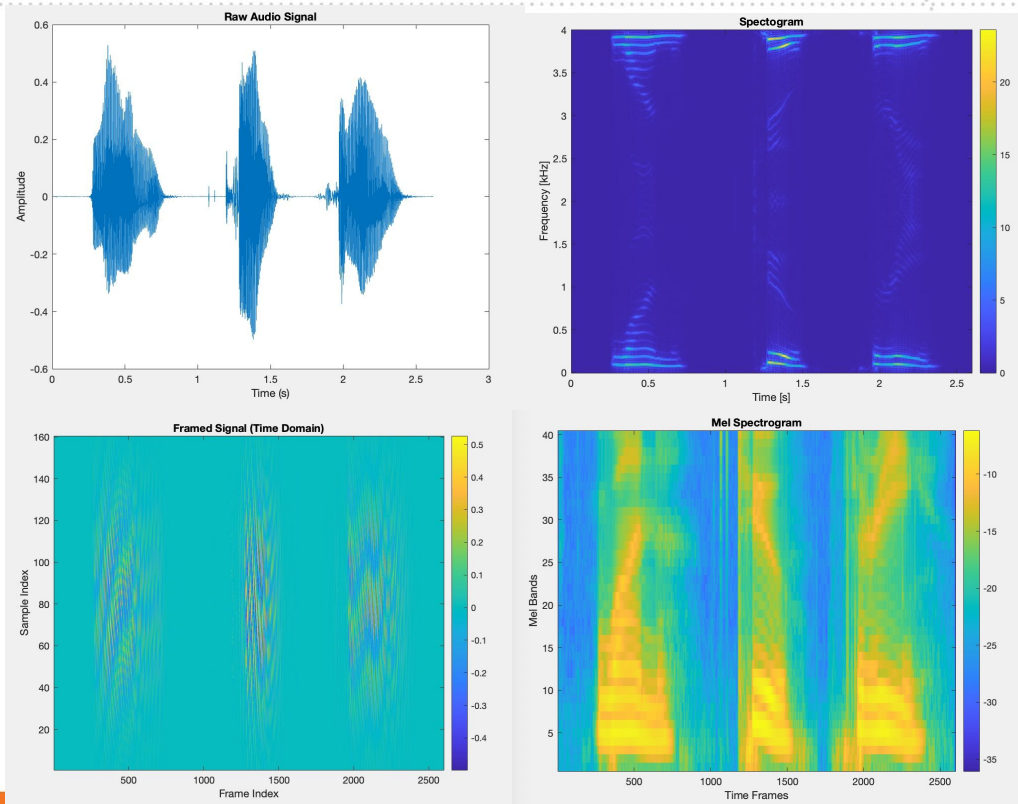
1. Converting to Tensor
2. Resampling to 16kHz
 - a. HomeGrown and Transfer model expects 16kHz
3. Convert to log mel spectrogram for homegrown



What is a Mel Spectrogram

A Mel Spectrogram is a specific graph showing frequencies based on the Mel scale:

1. Framing the Signal into a matrix of short segments of the audio file.
2. FFT is taken on windowed segments changing the signal into the frequency domain
 - a. Within this stage it is important to remember that when using the FFT the signal can and will alias.
3. The system is then converted into the Mel scale giving a much better image for learning models and mimics human hearing more effectively
 - a. This is done through the use of the Mel Filter Banks, which is a set of overlapping triangular filters.



Fine Tuned Feature Extraction

- Model: WavLM
- Input 16kHz audio sample
- Pass through WavLM feature extractor
- Pass through model
- Output: 256 dimension vector
- Dataset: Voxceleb

Homegrown Feature Extraction

- Architecture: Conv1d(128) -> Conv1d(128) -> Conv1d(256) -> Transformer Encoder -> Feature Vector
- Inputs: Log mel spectrogram
- Outputs: 256 dimension vector

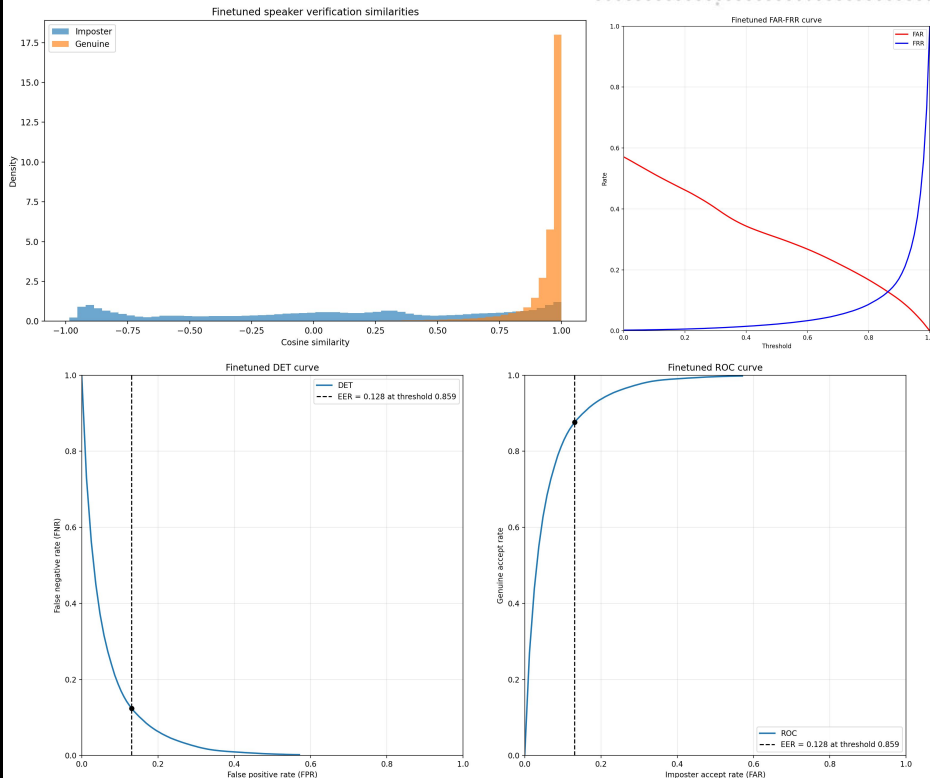
Performance

Fine Tuned:

Threshold: .859

Although in practice it seems to work better at a .95 threshold due to no speaking resulting in a high score

Similarity Score: Calculated using Cosine Similarity



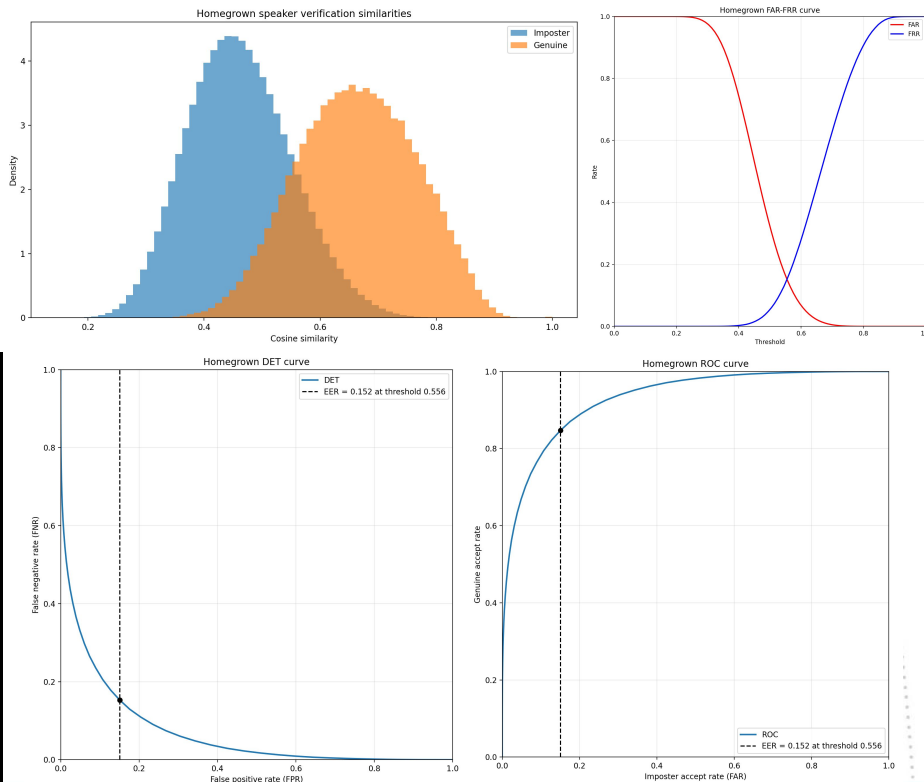
Performance

Home Grown:

Threshold: .556

Although in practice a threshold of around .7 works better due to no speaking resulting in a similarity score of .6

Similarity Score: Calculated using Cosine Similarity



Overall Performance

Our system seems to work relatively well, but there are some places for improvement:

1. Our system has a harder time differentiating between a male and a male than a male and a female
2. The fine tuned data set has a much higher similarity score threshold, but our homegrown systems seems to work better, resulting in less false positives.

Fun Application

Our system is also connected to feather esp32-s3, which will wait for a single to be sent over USB. When sent it will turn on a Motor based on if the user was in the database.

